Probabilistic Inventory models: Part 1

Fifty seventh Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



- **To begin with, one deterministic problem:** An industry requires 4,000 kg of raw material for manufacturing an item per year. The cost of placing an order is Rs. 35 and the cost of carrying inventory is 20 per cent of the investment in the inventories. The price is Rs. 12 per kg. Determine (a) optimal lot size, (b) optimal order cycle time, (c) minimum inventory variable cost (includes holding and set up cost but not the fixed purchase cost.)
- **Solution:** Given,

Annual demand, D = 4,000 kg per year; Ordering cost, C_o = Rs. 35 per order

Holding cost, C_h = 20% of the price/unit of raw material = Rs 12 × 0.2 = Rs 2.4 per kg per year

(a) Optimal lot size, Q^*

$$Q^* = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2\times 4000\times 35}{2.4}} = 341.56 \text{ kg per order}$$

(b) optimal order cycle time, t^*

$$t^* = \frac{Q^*}{D} = \frac{341.56}{4000} = 0.085 \text{ year}$$

(c) Per year minimum total inventory variable cost, T_{VC}

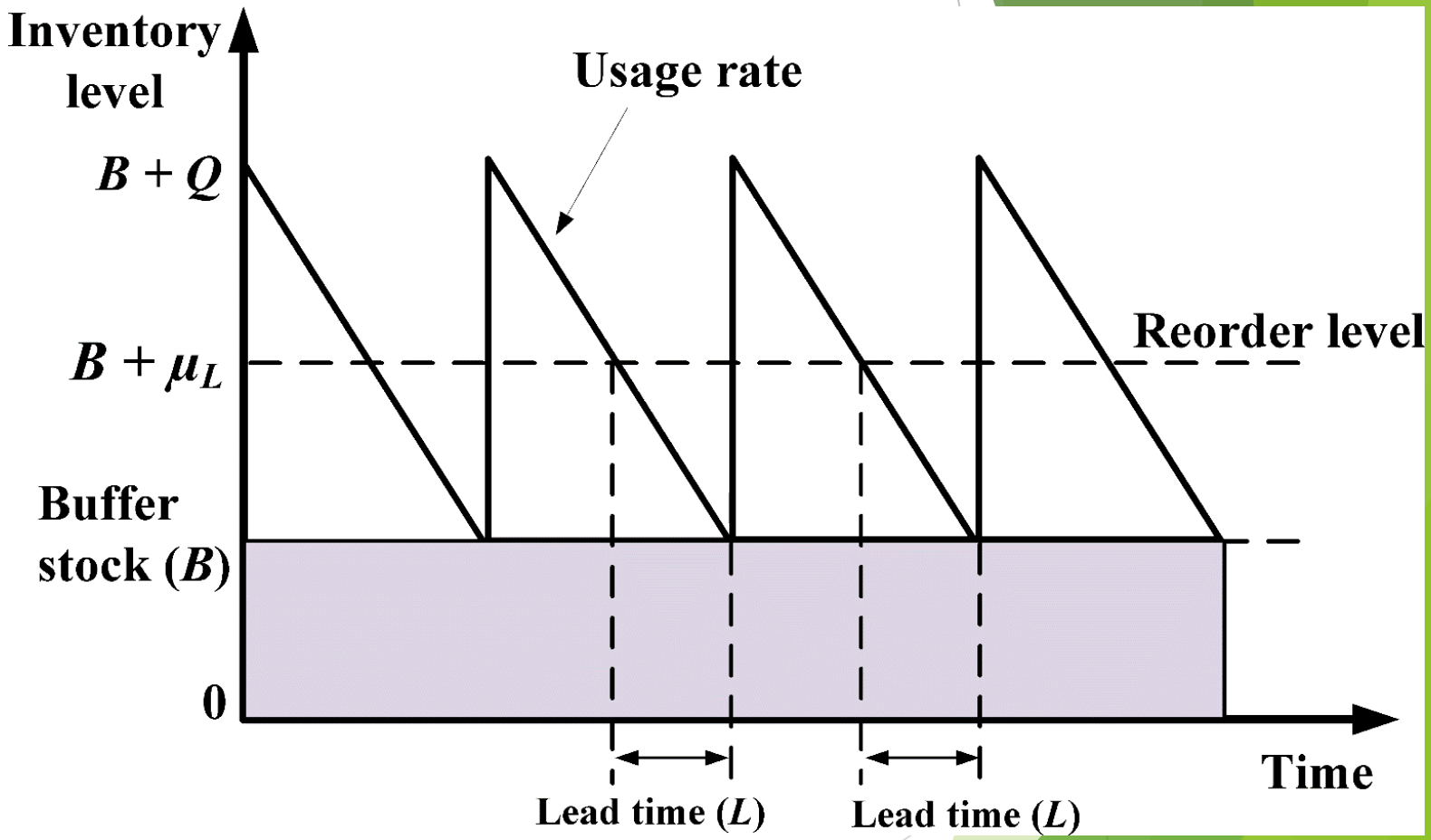
$$T_{VC} = \sqrt{2DC_oC_h} = \sqrt{2 \times 4000 \times 35 \times 2.4} = 819.756 \text{ per year}$$

Probabilistic EOQ model

- Probabilistic Economic Order Quantity (EOQ) model is an extension of the classical EOQ model that incorporates uncertainty in demand during the lead time.
- The classical EOQ model assumes deterministic and constant demand. However, the probabilistic version accounts for the random or variable nature of demand, which is more realistic in many practical scenarios.

- The critical period during inventory cycle occurs between placing and receiving orders. In this period, shortage of stock may occur.
- Hence, a buffer stock has to be maintained.

Figure shows the relationship between buffer stock or safety stock B and the parameters of deterministic EOQ model that includes Q, lead time L and average demand μ_L during lead time



- **Assumptions:**

- Demand per unit time is normal with mean *D* and standard deviation σ, *i.e.*, *N*(*D*,σ)
- Under this assumption, the demand during lead time *L* should be normal with mean $\mu_L = DL$ and standard deviation $\sigma_L = \sqrt{L\sigma^2}$
- The formula for σ assumes that L is an integer value

The size of the buffer *B* is determined such that the probability of shortage during *L* is at most α. Let x_L be the demand during lead time *L*, then

$$P\{x_L \geq B + \mu_L\} \leq \alpha$$

With $N(0, 1)$, $z = \dfrac{x_L - \mu_L}{\sigma_L}$, one obtains

$$P\left\{z \geq \frac{B}{\sigma_L}\right\} \leq \alpha$$

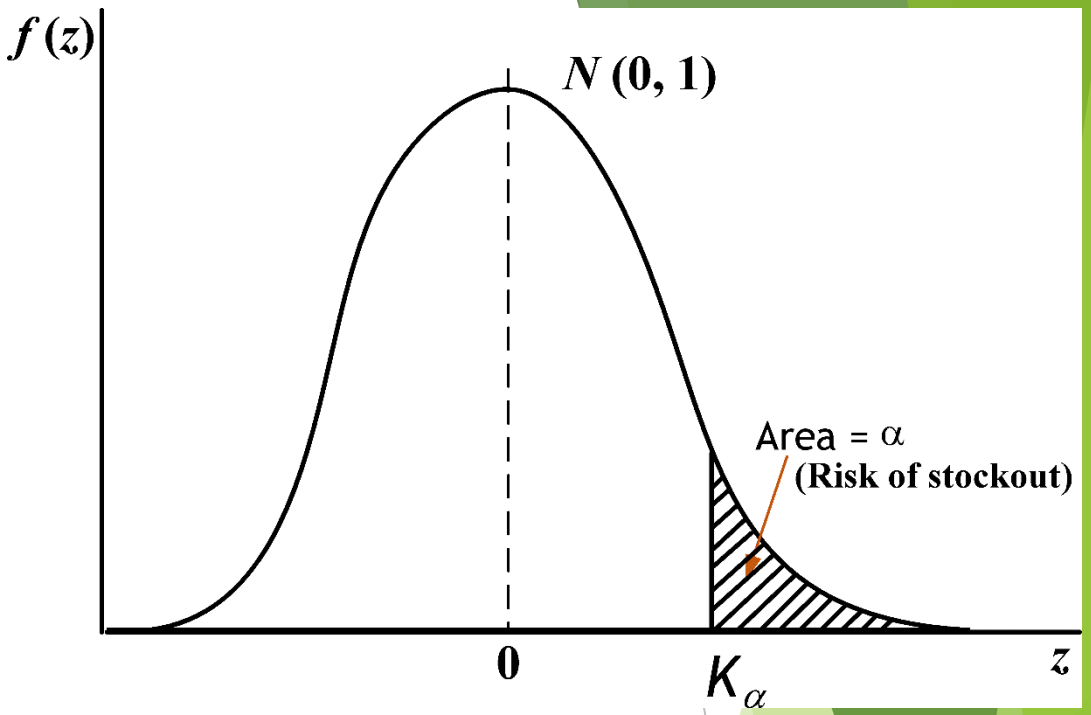


Figure defines the parameter K_α for

the standard normal distribution such that $P\{z \geq K_\alpha\} \leq \alpha$.

It follows that

$$B \geq \sigma_L K_\alpha$$

This amount provides the minimum value of B, i.e, the buffer stock

Question: The EOQ of a product is 1000 units. Assume that the daily demand is $N(100,10)$, i.e., D = 100 units and standard deviation σ = 10 units. Determine the buffer size, B, using α = 0.05. The effective lead time L is 4 days.

Solution:

$$\sigma_L = \sqrt{\sigma^2 L} = \sqrt{10^2 \times 4} = 20 \text{ units}$$

From standard normal table, the value of $K_{0.05}$ is 1.645.

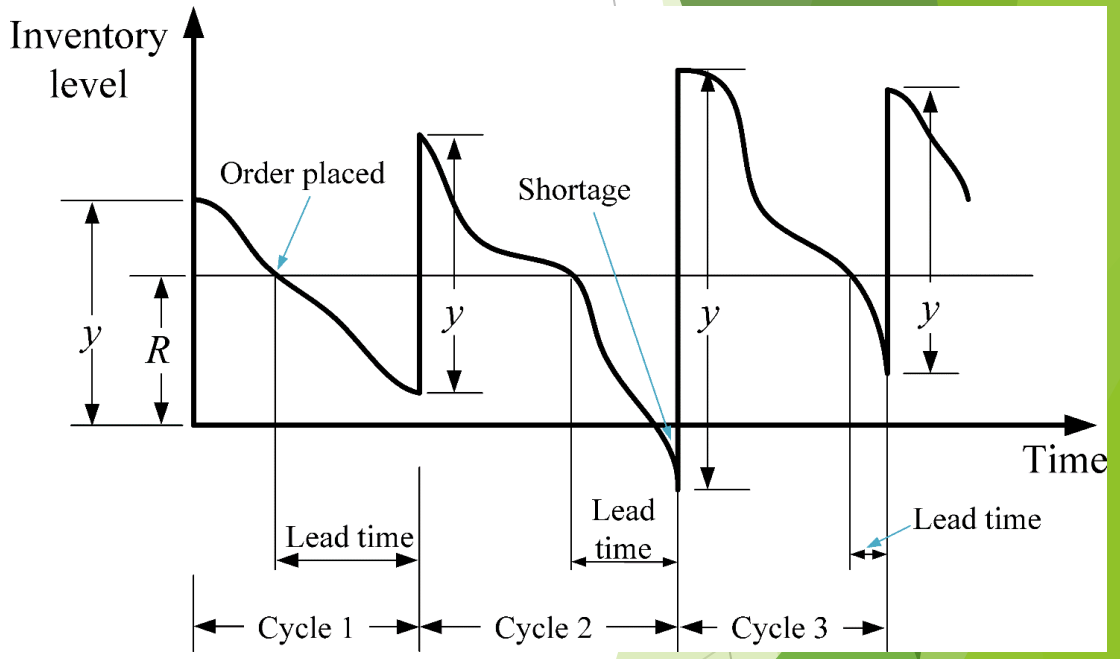
The buffer size is computed as

$$B \geq \sigma_L K_\alpha$$

or $B \geq 20 \times 1.645 = 33$ units

Probabilistic EOQ Model

- Figure depicts a typical change in inventory level with time.
- Shortage may or may not occur during (possibly random) lead times, as illustrated by cycles 1, 2 and 3, respectively.
- The policy calls for ordering the quantity *y* whenever the amount of inventory on hand drops to level *R*.
- In the deterministic case, the reorder level *R* is a function of the lead time between placing and receiving an order.
- The optimal values of *y* and *R* are determined by minimizing the expected sum of setup, holding, and shortage costs per unit time.



The model is based on three assumptions:

1. Unfilled demand during lead time is backlogged.
2. No more than one outstanding order is allowed.
3. The distribution of demand during lead time remains stationary with time.

To develop the total cost function per unit time, let

$f(x)$ = pdf of demand during lead time. Note: x is units of item.

D = Expected demand per unit time

h = Holding cost per inventory unit per unit time

p = Shortage cost per inventory unit

K = Setup cost per order

The elements of the cost function are now determined.

1. **Setup cost:** The approximate number of orders per unit time is D/y, so that the setup cost per unit time is approximately KD/y.

2. **Expected holding cost:** Given I is the average inventory level, the expected holding cost per unit time is hI. The average inventory level is computed as

$$I = \frac{\{y + E(R - x)\} + E(R - x)}{2} = \frac{y}{2} + R - E\{x\}$$

The formula averages the starting and ending expected inventories in a cycle: $y + E\{R-x\}$ and $E\{R-x\}$, respectively. As approximation, the expression ignores the case where $R-E\{x\}$ may be negative.

3. **Expected shortage cost:** Shortage occurs when x > R. Its expected value per cycle is computed as

$$S = \int_{R}^{\infty} (x - R) f(x)\,dx$$

Because *p* is assumed to be proportional to the shortage quantity only, the expected shortage cost per cycle is *pS*, and, based on *D*/*y* cycles per unit time, the shortage cost per unit time is $\frac{pDS}{y}$

The resulting total cost function per unit time is

$$T_c(y, R) = \frac{DK}{y} + h\left(\frac{y}{2} + R - E\{x\}\right) + \frac{PD}{y}\int_{R}^{\infty} (x - R) f(x)\,dx$$

The optimal values, y^* and R^*, are determined from

$$\frac{\partial T_c}{\partial y} = -\left(\frac{DK}{y^2}\right) + \frac{h}{2} - \frac{pDS}{y^2} = 0$$

$$\frac{\partial T_c}{\partial R} = h - \left(\frac{pD}{y}\right)\int_R^\infty f(x)\mathrm{d}x = 0$$

These two equations yield

$$y^* = \sqrt{\frac{2D(K+pS)}{h}}$$

$$\int_{R^*}^\infty f(x)\,\mathrm{d}x = \frac{hy^*}{pD}$$

The optimal values of y^* and R^* cannot be determined in closed forms.

The algorithm converges in a finite number of iterations, provided a feasible solution exists.

For $R = 0$, previous two equations yield

$$\hat{y} = \sqrt{\frac{2D(K + pS)}{h}}$$

$$\tilde{y} = \frac{PD}{h}$$

Unique optimal values of y and R exist when $\tilde{y} \geq \hat{y}$. The smallest value of y^* is

$$\sqrt{\frac{2KD}{h}}$$

It occurs when $S = 0$.

The steps of the algorithm are

Step 1: Use the initial solution

$$y_1 = y^* = \sqrt{\frac{2KD}{h}}$$

Let $R_0 = 0$. Set $i = 1$, and go step i

Step 2: Use y_i to determine R_i . If $R_i \approx R_{i-1}$, stop; the optimal solution is $y^* = y_i$, and $R^* = R$;. Otherwise, use R_i to compute y_i. Set $i = i + 1$, and repeat step i.